



Velammal College of Engineering and Technology, Madurai
(Autonomous)
Department of Computer Science and Engineering

Course Code-Title	21CS302 / COMPUTER NETWORKS				
Course Component	Professional Core				
Contact Hours	Lecture	Tutorial	Practical	Total Hours	Credit
	3	0	2	75	4
Course Assessment methods	Continuous			Semester-End Assessment	
	Internal Assessment Tests Model Lab Assignments			Student Course Exit Survey	
Prerequisite Courses:	Knowledge in Basic knowledge in computers.				
Course Objective	Students should be made to - To describe the Network Architecture and the performance metrics of switched networks. - To demonstrate the various Link layer services. - To explain the concepts of subnetting and routing mechanisms. - To illustrate the process-to-process delivery models and congestion control principles. - To summarize the services of various protocols in Application layer.				
TOPICS TO BE COVERED					
UNIT I INTRODUCTION AND PHYSICAL LAYER				9	
Networks – Network Types – Protocol Layering – TCP/IP Protocol suite – OSI Model – Physical Layer: Performance – Transmission media – Switching – Circuit-switched Networks – Packet Switching.					
UNIT II DATA-LINK LAYER				9	
Introduction – Link-Layer Addressing – Error Detection and Correction– DLC Services-Framing –Data Link Layer Protocols – HDLC – PPP - Media Access Control –Random access- Wired LANs: Ethernet -Wireless LANs –IEEE 802.11-Connecting Devices.					
UNIT III NETWORK LAYER				9	
Network Layer Services– IPV4 Addresses-Classful Addressing-Classless Addressing-Dynamic Host Configuration Protocol (DHCP)- Network Layer Protocols: IP, ICMP v4 – Unicast Routing Algorithms and Protocols- IPV6 Addressing.					
UNIT IV TRANSPORT LAYER				9	
Introduction – Transport Layer Protocols – Services – Port Numbers – User Datagram Protocol – Transmission Control Protocol – SCTP.					
UNIT V APPLICATION LAYER				9	
World Wide Web and HyperText Transfer Protocol – File Transfer Protocol – Electronic Mail –Telnet –Secure Shell – Domain Name System – Simple Network Management Protocol.					
				45 PERIODS	
PRACTICAL EXERCISES					30 PERIODS
1. Make use of various networking commands like tcpdump, netstat, ifconfig, nslookup and traceroute. Capture ping and traceroute PDUs using a network protocol analyzer and examine. 2. Design a topology using PCs and Switch with configuration of IP address and Observe the flow of data from host to host by creating network traffic. 3. Create a Network scenario and examine dynamically learning configured Switch MAC address table and ARP Cache table using simulation tool. 4. Simulation of Error correction and detection techniques.					

5. Create a Network Scenario and assign subnet IP Addresses to various Network Devices and Verify the Connectivity using simulation tool. 6. Create a Network scenario with multiple routers and configure using RIP Routing in simulation tool. 7. Create a Network scenario with multiple routers and configure using OSPF Routing in simulation tool. 8. Create a Network scenario and generate the network traffic to examine the TCP/UDP communication using Simulation tool. 9. Implement the applications using TCP /UDP sockets like: • Chat • File transfer 10. Setting up DNS, HTTP, DHCP and E-mail server using simulation tool.		
Course Outcomes	CO1: Make use of evaluation metrics to measure the performance of packet switched network.(K3) CO2: Utilize the Link layer services for various IEEE standards.(K3) CO3: Experiment with subnetting to optimize network configuration and various routing algorithms for unicast routing.(K3) CO4: Choose protocols for Process to Process communication in various application.(K3) CO5: Utilize application layer protocols for real time Scenario.(K3)	
TEXT BOOKS: 1. Behrouz A. Forouzan, “Data Communications and Networking”, 5th Edition, Tata McGraw Hill, 2017. 2. William Stallings, “Data and Computer Communication”, 10th Edition, Pearson Education, 2022. 3. Larry L. Peterson, Bruce S. Davie, “Computer Networks: A Systems Approach”, 6th Edition, Morgan Kaufmann Publishers Inc., 2017.		
REFERENCES: 1. Nader F. Mir, “Computer and Communication Networks”, 2nd Edition, Prentice Hall, 2015. 2. James F. Kurose, Keith W. Ross, “Computer Networking, A Top-Down Approach Featuring the Internet”, 8th Edition, Pearson Education, 2022 3. Mulayam Singh ,“CISCO PACKET TRACER LABS: Best practice of configuring or troubleshooting Network”, 1st Edition ,BookRix,2019.		
WEB MATERIALS: W1.https://www.youtube.com/watch?v=-6Uoku-M6oY W2.http://www.dauniv.ac.in/downloads/EmbsysRevEd_PPTs/Chap_3Lesson25EmsysNew.pdf W3.https://learn.sparkfun.com/tutorials/bluetooth-basics/all W4.https://www.cisco.com/c/en/us/support/docs/ip/routing-information-protocol-rip/13788-3.pdf W5..https://nptel.ac.in		

Course In-Charge

Course Coordinator

Module Coordinator

HoD-CSE